Chaklam Silpasuwanchai, Todsavad Tangtortan Updated 21 March 2026

AT82.05 Artificial Intelligence: Natural Language Understanding (NLU)

A7: MCP-Server, AI Agent, and External Tool Integration

In this assignment, you will build an integrated AI Agent ecosystem using the Model Context Protocol (MCP). The goal is to move beyond simple chat and create an agent capable of managing real-world schedules and communicating via popular messaging platforms, demonstrating practical Natural Language Understanding (NLU)[1]. In this exercise, we will deploy **n8n locally** using Docker and expose it to the internet using **ngrok** to handle external webhooks and API communications.

Task 1. MCP Infrastructure & Server Setup (3 Points)

1) **Server Deployment:** Run your n8n environment using Docker. Use ngrok or a similar tunnel to ensure your Production URL is accessible from the internet. (0.5 point)
2) **MCP Server Workflow:** Create an n8n workflow acting as an **MCP Server**.
   - Implement an **MCP Trigger** with at least three internal tools (e.g., `Calculator`, `Date/Time`, `Text Formatter`).
   - Verify that the server is active and the tools are discoverable. (1 point)
3) **AI Agent Client:** Create a separate **AI Agent** workflow.
   - Configure the Agent with a **Free Model API** (e.g., Groq[2] or Typhoon[3]) and use **Window Buffer Memory** to maintain context.
   - Connect the **MCP Client (Tool)** to your **MCP Server**'s Production URL.
   - **Verification:** Open the n8n chat and confirm the agent can use MCP tools. (1.5 points)

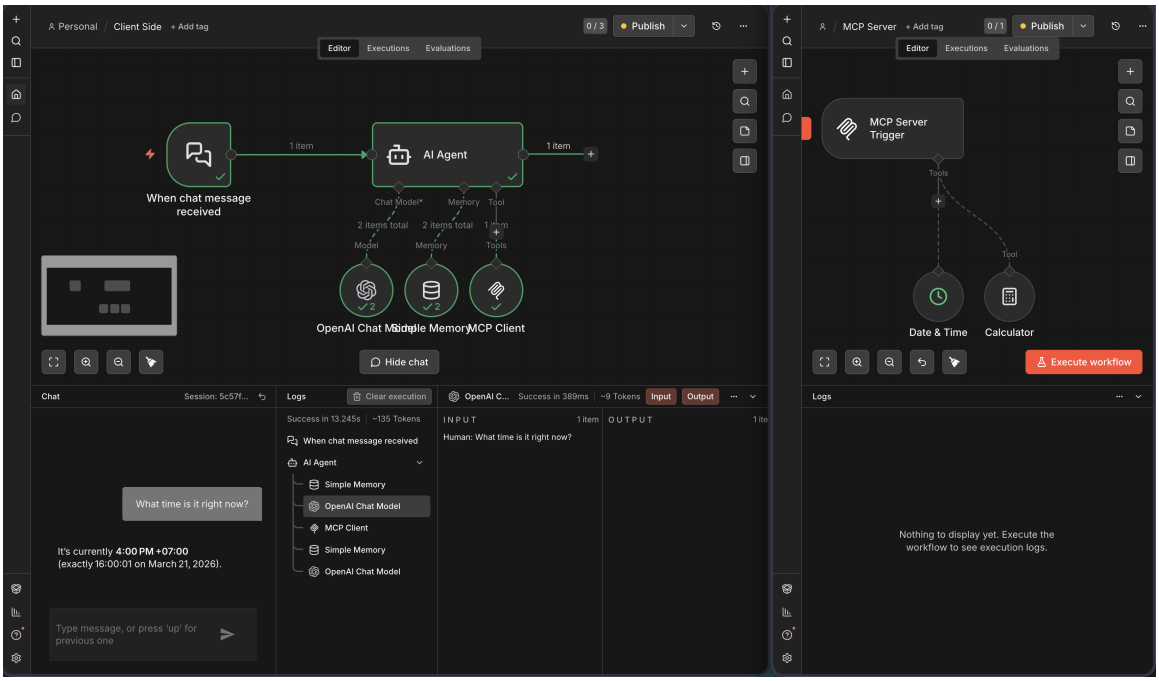

FIGURE 1. Example of an n8n MCP Server and Agent workflow configuration.

[1]For environment setup and footprints, refer to: https://github.com/chaklam-silpasuwanchai/Python-fo-Natural-Language-Processing/tree/main/Code/11%20-%20Agentic%20AI/local-n8n

[2]https://console.groq.com/

[3]https://playground.opentyphoon.ai/

Task 2. Telegram & Google Calendar Integration (3 Points)

1) **Telegram Bot API:** Connect your AI Agent to **Telegram**. Use a **Telegram Trigger** so the agent can receive and reply to messages directly on the Telegram app. (0.5 point)
2) **Google Calendar Tool:** Integrate the **Google Calendar Tool** into your Agent. The agent must have the capability to create, read, and manage events. (0.5 point)
3) **Automated Project Scheduling:** Test your agent by giving it a command via Telegram to create a schedule for a project. The agent should create the following four events in your Google Calendar:
   - **1st Phase:** Literature Review
   - **2nd Phase:** Project Proposal
   - **3rd Phase:** Update Progress
   - **4th Phase:** Final (Presentation)

   (1 point)
4) **Interaction Verification:** Ask the chatbot (via Telegram) about the project phases or to verify if they are on the calendar. Provide screenshots of:
   - The conversation in Telegram where you command the agent.
   - The resulting events appearing in your **Google Calendar**. (1 point)